EXP NO (1) FREQUENCY RESPONSE OF COMMON
27/1/26 EMITTER AMPLIFIER

AIM:

To design a single-stage common-emitter amplifier using a transistor, simulate the circuit in LT spice, analyze its frequency response, and calculate gain and bandwidth.

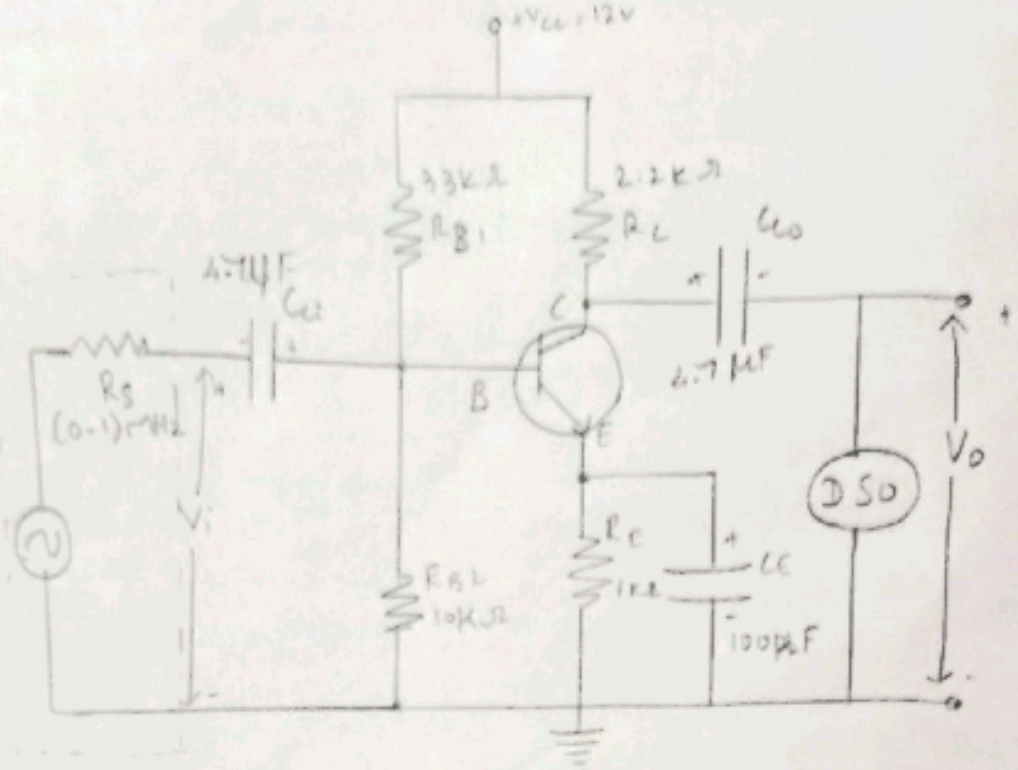
SOFTWARE REQUIRED:

LT spice simulator.

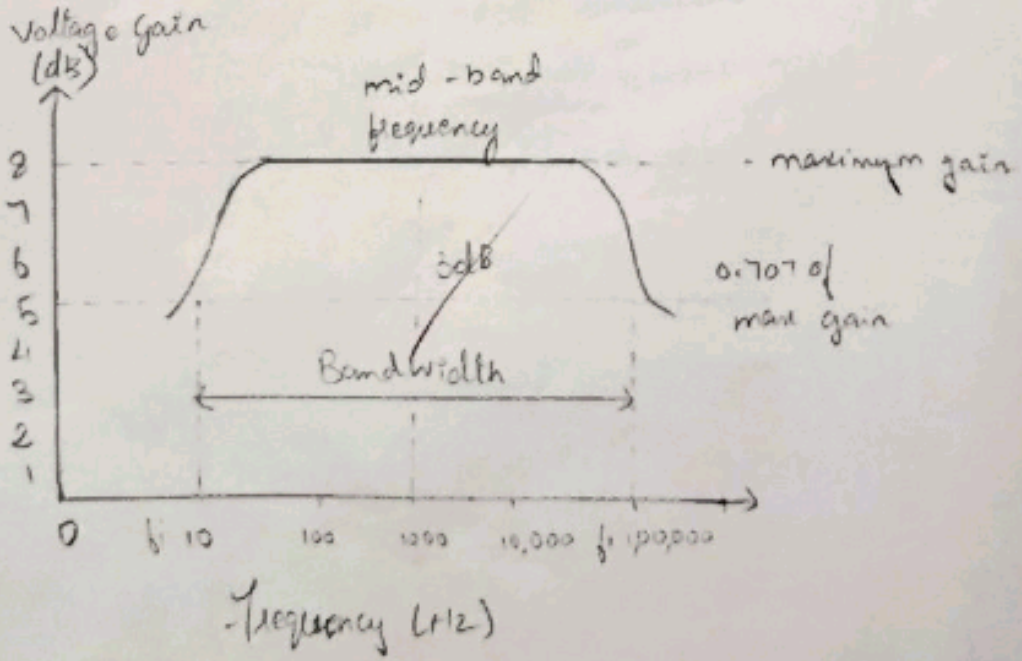
PROCEDURE:

Connect the circuit as per the circuit diagram using LT spice simulator.

CIRCUIT DIAGRAM:



MODEL GRAPH



TABULATION:

$V_{in} = 1V$

S.NO	Frequency (Hz)	Vo(V)	Vo/Vin(V)	GAIN = 20log (Vo/Vin) (dB)
1	10	0.5	0.5	-6.02
2	50	0.7	0.7	-3.10
3	100	1.0	1.0	0
4	1k	2.5	2.5	7.96
5	10k	2.8	2.8	8.94
6	50k	2.7	2.7	8.63
7	100K	2.0	2.0	6.02
8	200K.	1.2	1.2	1.58

Calculation:-

1) Gain Calculation: $gain(dB) = 20\log\left(\frac{V_o}{V_{in}}\right)$, $V_{in} = 1V$

$gain(dB) = 20\log V_o$

For frequency = 1KHz, $V_o = 2.5V$

$Gain = 20\log V_o = 20\log(2.5)$, $= 20 \times 0.397 = 7.96dB$.

For frequency = 10Hz, $V_o = 0.5V$

$Gain = 20\log V_o = 20\log(0.5) = 20 \times (-0.301) = -6.02dB$

Bandwidth: $= f_H - f_L$.

$f_H = 100kHz$; $f_L = 50Hz$

$BW = 100000 - 50 = 99.950N$.

RESULT:-

The Common Emitter Amplifier was constructed and the following results were observed.

1. Maximum signal handling capacity .VO≃2.8 V
2. Q-point : ICQ ..... mA
   VCEQ = .6.. V
3. Gain of the amplifier .8-9 dB
4. Bandwidth of the amplifier 100K. Hz.
5. Gain- Bandwidth Product ..8.X10^5.